

Peripheral blood transcriptomic signatures of Alzheimer's disease show cohort-dependent reproducibility and limited diagnostic transferability

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Abstract

Background

Peripheral blood transcriptomic profiles have been used to propose diagnostic signatures for Alzheimer's disease (AD), but their reproducibility and transferability across cohorts and platforms remain uncertain. Single-cohort discovery studies can identify strong apparent signals that do not necessarily generalize to independent datasets.

Methods

We performed a staged validation audit of public peripheral blood transcriptomic datasets. GSE63060 was used as the discovery cohort, GSE63061 as a related validation cohort, and GSE85426 plus GSE140829 as independent cross-platform validation cohorts. Differential-expression candidates were frozen in GSE63060 using linear modeling adjusted for age and sex. Validation analyses quantified candidate-gene mapping, direction concordance, nominal and FDR-level replication, gene-level oriented AUC, signature/model AUC, stable and unstable transportability classes, and GO BP, KEGG, and Reactome enrichment. GSE85426 and GSE140829 were not used to optimize candidate selection or tune diagnostic models.

Results

The discovery-derived candidate set showed strong related-cohort reproducibility in GSE63061, with 476/476 mapped genes, direction concordance of 0.989, 448 nominally concordant genes,

395 FDR-concordant genes, median oriented gene-level AUC of 0.637, and best model AUC of 0.784. In contrast, cross-platform transferability was substantially weaker. GSE85426 mapped 451/476 candidates but showed direction concordance of 0.749, 35 nominally concordant genes, no FDR-concordant genes, median oriented AUC of 0.537, and best model AUC of 0.587. GSE140829 mapped 460/476 candidates and showed direction concordance of 0.898, 122 nominally concordant genes, 5 FDR-concordant genes, median oriented AUC of 0.534, and best model AUC of 0.569. Strict stable genes numbered 103, whereas 235 genes were classified as unstable. Stable genes were enriched for translation, ribosomal, and RNA-processing pathways, while unstable genes showed no FDR-significant GO BP, KEGG, or Reactome enrichment.

Conclusions

AD-related peripheral blood transcriptional signals are detectable and can reproduce strongly in a related cohort, but diagnostic transferability is limited across independent platforms and cohorts. These findings support a cautious interpretation of public blood transcriptomic signatures and reinforce the need for strict external validation before biomarker claims.

Keywords: Alzheimer's disease; peripheral blood; transcriptomics; external validation; transportability; diagnostic transferability; public datasets; pathway enrichment.

Background

Alzheimer's disease (AD) is the most common cause of dementia and a major contributor to disability and care burden in ageing populations. The World Health Organization estimates that 57 million people were living with dementia worldwide in 2021 and that AD may account for 60-70% of dementia cases [1]. Contemporary research criteria increasingly define AD through biological processes, particularly amyloid, tau, and neurodegeneration biomarkers, rather than through clinical symptoms alone [2]. This biomarker shift has strengthened disease classification, but many established biomarkers still rely on cerebrospinal fluid, positron-emission tomography, or specialized assays that are not equally accessible in all settings.

Blood-based molecular readouts are therefore attractive for AD research because peripheral sampling is comparatively scalable, repeatable, and compatible with large cohorts. Whole-blood or peripheral-blood transcriptomic profiles are one such readout. They may capture immune activation, inflammatory tone, metabolic state, cellular composition, and systemic physiological changes that accompany neurodegenerative disease. Earlier blood biomarker reviews emphasized both the promise of minimally invasive assays and the difficulty of achieving reproducible performance across cohorts [3,4]. Blood gene-expression studies have reported AD-associated classifiers or candidate markers using whole-blood or leukocyte transcriptomic profiles, including early microarray classifiers, leukocyte biomarker work, and AddNeuroMed-based signatures for cognitive impairment or early AD detection [5-8]. These findings helped establish peripheral blood transcriptomics as a plausible exploratory biomarker domain. At the same time, blood transcriptomic signatures are exposed to multiple sources of instability. Measured expression can vary with blood-cell composition, age, sex, medication exposure, comorbidity, RNA handling, microarray chemistry, preprocessing, probe annotation, and cohort recruitment. These factors are not merely technical nuisances: they can create signatures that appear strong in one cohort yet fail to retain direction, statistical support, or diagnostic discrimination elsewhere. For this reason, a clinically oriented biomarker claim requires strict external validation rather than only internal resampling or validation in a closely related dataset. Public transcriptomic repositories make this problem testable. The Gene Expression Omnibus (GEO) contains multiple peripheral-blood AD microarray datasets with AD, mild cognitive impairment (MCI), and control samples. GSE63060 and GSE63061 are AddNeuroMed blood-expression datasets generated in related study contexts but on different Illumina HumanHT-12 platform versions [9,10]. GSE85426 is an independent Agilent blood microarray dataset with 90 AD cases and 90 non-demented controls [11]. GSE140829 is a large peripheral-blood expression dataset spanning AD, MCI, controls, and other dementia disorders, with a reported innate immune inflammatory signal across neurodegenerative conditions [12]. Together, these

78 datasets permit a staged assessment of related-cohort reproducibility and cross-platform
79 transferability.

80 Many public-data biomarker studies emphasize the best-performing model after feature filtering,
81 threshold selection, or repeated dataset reuse. That practice can lead to optimistic conclusions
82 when external datasets influence candidate selection or model tuning. Reporting guidance for
83 prediction models, including TRIPOD, stresses transparent separation of model development
84 and validation, because diagnostic performance estimates are difficult to interpret when the
85 validation role is blurred [13]. In AD blood transcriptomics, the distinction is especially important:
86 a signature that performs well in a related AddNeuroMed-like cohort may still be insufficiently
87 transferable to an independent cohort with different platform chemistry or recruitment.

88 We therefore designed this study as a transportability audit rather than as a search for a high-
89 performance diagnostic classifier. GSE63060 was used for discovery, GSE63061 for related-
90 cohort validation, and GSE85426 plus GSE140829 for independent cross-platform validation.

91 The central question was not whether a model could be tuned until it exceeded an AUC
92 threshold in a known validation set, but whether a discovery-derived blood transcriptional signal
93 retained direction, statistical support, and diagnostic discrimination across increasingly
94 independent validation contexts.

95 The study had four objectives. First, we quantified candidate-gene reproducibility across
96 validation cohorts using mapping rate, direction concordance, nominal replication, FDR-level
97 replication, and gene-level oriented AUC. Second, we evaluated whether simple and penalized
98 diagnostic models selected without using final validation labels retained performance across
99 platforms. Third, we separated stable transportable genes from unstable genes that reversed
100 direction or lost cross-platform signal. Fourth, we tested whether stable and unstable gene sets
101 had coherent pathway context. This framing allows external-validation failure to be treated as
102 the main empirical finding, not as an inconvenient negative result.

Methods

Study design

This was a retrospective public-data bioinformatics validation study. The analysis followed a staged design with prespecified dataset roles. GSE63060 was used as the discovery cohort. GSE63061 was used as a related external validation cohort because it comes from the same broad AddNeuroMed context but a different batch/platform. GSE85426 was used as a cross-platform stress-test cohort. GSE140829 was introduced after initial model-performance concerns as a large independent cross-platform validation cohort. The final Route B manuscript was triggered after diagnostic-model rescue attempts failed to achieve $AUC \geq 0.70$ in independent cross-platform validation.

All datasets were publicly available and de-identified. No new human-subject recruitment, intervention, or sample collection was performed.

Datasets and sample inclusion

The primary discovery dataset was GSE63060, an AddNeuroMed peripheral blood microarray dataset measured on GPL6947 Illumina HumanHT-12 V3.0 expression beadchip [9]. The locked AD-versus-control analysis included 145 AD cases and 104 controls.

The related validation dataset was GSE63061, an AddNeuroMed peripheral blood microarray dataset measured on GPL10558 Illumina HumanHT-12 V4.0 expression beadchip [10]. The locked AD-versus-control analysis included 139 AD cases and 134 controls.

The first cross-platform validation dataset was GSE85426, a peripheral blood microarray dataset measured on GPL14550 Agilent-028004 SurePrint G3 Human GE 8x60K Microarray [11]. The locked AD-versus-control analysis included 90 AD cases and 90 controls.

The second cross-platform validation dataset was GSE140829, a large peripheral blood expression dataset measured on GPL15988 HumanHT-12 v4 Expression BeadChip [12]. The analyzable normalized expression subset contained 198 AD cases, 229 controls, and 124 MCI

samples. AD-versus-control discrimination was the primary endpoint, while MCI score placement was retained as exploratory context only.

Data freeze and quality control

Raw and normalized GEO files were downloaded and recorded with source URLs, file sizes, and SHA-256 checksums. Sample metadata were parsed from the GEO series matrix files, and AD/control inclusion was locked before modeling. Dataset-level quality control checked feature counts, sample counts, missingness, expression distributions, and PCA structure. Processed expression matrices were stored as RDS files and were not redefined during downstream modeling.

Discovery differential-expression analysis

Differential expression was performed in GSE63060 only. Linear models were fitted using limma with diagnosis as the main coefficient and age and sex as covariates when available [14]. Candidate genes were selected using Benjamini-Hochberg adjusted $P < 0.05$ and absolute log fold-change ≥ 0.20 . Probe annotation used GEO platform files where available. When a probe annotation field contained multiple symbols or Entrez identifiers, the first non-empty token separated by common GEO delimiters was used. Candidate genes were collapsed to unique gene symbols by retaining the probe with the lowest adjusted P value and, if needed, the larger absolute log fold-change. The main discovery output contained 623 candidate probes, 546 annotated candidate probes, and 476 unique genes. GSE63061, GSE85426, and GSE140829 were not used to select discovery candidates.

Candidate-gene mapping and external validation

Discovery candidates were mapped into each validation dataset at the Entrez gene or gene-symbol level depending on the available platform annotation. For GSE63061, GPL10558 annotation was used. For GSE85426, row-name Entrez suffixes were used because a standard GEO platform annotation file was not available in the analysis workflow. For GSE140829, the normalized expression matrix used gene symbols as row identifiers. In validation datasets with

multiple probes mapping to the same Entrez gene, probe-level variance was calculated on the normalized expression matrix and the highest-variance probe was retained as the gene-level representative. Expression columns were matched to locked sample metadata before modeling, and scripts stopped when metadata and matrix order did not align.

For each validation cohort, gene-level validation models estimated the AD-versus-control log fold-change while adjusting for available covariates. Direction concordance was defined as agreement between the sign of the GSE63060 discovery log fold-change and the validation log fold-change. Nominal concordance required direction concordance and validation $P < 0.05$. FDR concordance required direction concordance and validation adjusted $P < 0.05$. Gene-level oriented AUC was calculated using the validation expression value multiplied by the sign of the discovery log fold-change, so that higher oriented values consistently represented the discovery AD direction.

Signature and diagnostic-model evaluation

The main modeling analyses used frozen discovery and validation roles. GSE63060 was used for training, GSE63061 for model selection or development evaluation, and GSE85426 for final stress testing. Primary models included a signed mean gene-only signature and an integrated penalized model including gene features plus age and sex where available. Gene-expression inputs were row-wise z-scored within each dataset before signature scoring or penalized modeling. Age was standardized using the GSE63060 training mean and standard deviation, and sex was encoded as a binary male indicator.

Penalized models were fitted with glmnet binomial models [15]. Candidate model grids included ranked feature sets of 5, 10, 15, 20, 30, 50, 100, or all mapped candidate genes; alpha values of 0, 0.5, or 1; lambda.min or lambda.1se selection; and gene-only or gene-plus-age/sex covariate modes. Five-fold cross-validation used AUC as the internal selection metric. glmnet standardization was disabled because gene features had already been z-scored and age had been standardized. A train-development refit model was also evaluated by refitting on

GSE63060 plus GSE63061 without using GSE85426 or GSE140829 labels for tuning. After GSE140829 was frozen as a new large validation cohort, the same frozen or pre-registered models were evaluated in GSE140829 without label-driven optimization. The prespecified diagnostic-performance threshold was $AUC \geq 0.70$ in an independent validation cohort. This was a pragmatic study gate used to prevent weak cross-platform discrimination from being interpreted as diagnostic evidence; it was not defined as a threshold for clinical adequacy. Models that did not meet this threshold were not interpreted as supporting clinical use. Because no model achieved this gate in independent cross-platform validation, calibration, sensitivity, specificity, and decision-curve analyses were not emphasized as clinical performance summaries.

Stable and unstable transportability classes

Route B stable and unstable gene sets were constructed after the diagnostic-model claim was downgraded. The strict stable set required nominal concordance in GSE63061, direction concordance in all mapped cross-platform validation cohorts, and nominal concordance in at least one cross-platform cohort. A directional stable sensitivity set required direction concordance in GSE63061 and all mapped cross-platform cohorts. The unstable set required nominal concordance in GSE63061 but failure to meet the strict stable definition, together with either cross-platform direction reversal or cross-platform signal loss. Signal loss was defined as no nominal cross-platform support and maximum cross-platform oriented $AUC < 0.55$.

Pathway enrichment

Pathway over-representation analysis was performed separately for strict stable, directional stable sensitivity, and unstable gene sets. The main background was the full S3 candidate gene universe, because the biological question was whether transportable genes differed from the pool of discovery-derived AD blood transcriptional candidates. GO Biological Process, KEGG, and Reactome analyses were performed with Benjamini-Hochberg FDR control using clusterProfiler and ReactomePA, with Reactome pathway definitions interpreted against the

Reactome Pathway Knowledgebase [17-20]. Enrichment results were used as biological context and not as causal evidence.

Statistical analysis and reproducibility

AUC values were calculated using rank-based or ROC-based methods depending on the analysis stage; ROC analyses used pROC where applicable [16]. Differential-expression P values were adjusted by the Benjamini-Hochberg method. Analyses were run in R 4.5.3 with data.table 1.17.8, limma 3.66.0, ggplot2 4.0.3, glmnet 4.1.10, pROC 1.19.0.1, clusterProfiler 4.18.4, org.Hs.eg.db 3.22.0, ReactomePA 1.54.0, AnnotationDbi 1.72.0, GO.db 3.22.0, KEGGREST 1.50.0, patchwork 1.3.2, and png 0.1.9.

The executable analysis chain is stored in the project scripts/ directory. The staged scripts were freeze_s1_geo_metadata.py for GEO source freeze and checksums; s2_preprocess_qc.R for preprocessing and quality control; s3_limma_deg.R for discovery differential expression; s4_external_validation.R for initial external validation; s5_model_optimization.R, s5_gse85426_posthoc_upper_bound.R, and s5_train_dev_refit_final_validation.R for model rescue and frozen refit evaluation; s6_risk_route_dataset_scan.R and s7_route_a_gse140829_validation.R for additional validation-route assessment; s8_route_b_b1_evidence_consolidation.R and s9_route_b_b2_stable_unstable_gene_sets.R for Route B evidence consolidation and stable/unstable gene-set construction; s10_route_b_b3_pathway_enrichment.R for pathway enrichment; and s11_route_b_b4_transportability_audit.R for transportability-audit figures and tables. Main outputs are stored in results/, figures in figures/route_b/, and audit reports in audit/.

Results

Dataset roles were separated to prevent validation leakage

The final Route B analysis used four datasets with distinct roles. GSE63060 was the discovery cohort with 145 AD cases and 104 controls. GSE63061 was the related validation cohort with 139 AD cases and 134 controls. GSE85426 was the cross-platform stress-test cohort with 90

AD cases and 90 controls. GSE140829 was the large independent cross-platform validation cohort with 198 AD cases, 229 controls, and 124 MCI samples in the analyzable normalized expression subset.

This design separated discovery, related-cohort validation, and cross-platform validation.

Candidate selection was restricted to GSE63060, and the cross-platform datasets were not used to tune feature thresholds or optimize models. The staged design is summarized in Figure 1.

Discovery-derived candidates replicated strongly in the related validation cohort

The GSE63060 discovery analysis identified 476 unique candidate genes after annotation and gene-level consolidation. In GSE63061, all 476 candidates were mapped. Direction concordance was 0.989, with 471 genes showing the same AD direction as in the discovery cohort. Nominal concordance was observed for 448 genes, and 395 genes remained concordant at $FDR < 0.05$. The median oriented gene-level AUC was 0.637, and the maximum oriented gene-level AUC was 0.761.

Model-level performance in the related validation context was also high. The primary gene-only model reached AUC 0.781 in GSE63061, and the primary integrated model reached AUC 0.784. These results indicate that the discovery-derived signal was not random within the related AddNeuroMed-like validation context.

Cross-platform validation showed substantial degradation

The same candidate set transferred less well to GSE85426. Of 476 discovery-derived genes, 451 were mapped. Direction concordance remained above chance at 0.749, but only 35 genes were nominally concordant and none were FDR-concordant. The median oriented gene-level AUC was 0.537, close to a weak discrimination range, although the maximum oriented gene-level AUC reached 0.652.

GSE140829 showed a different but still limited pattern. Mapping was high, with 460/476 genes available. Direction concordance was 0.898, higher than in GSE85426, but FDR-level

replication remained limited, with 122 nominally concordant genes and only 5 FDR-concordant genes. The median oriented gene-level AUC was 0.534 and the maximum oriented gene-level AUC was 0.604. Thus, high direction concordance in GSE140829 did not translate into strong gene-level discrimination.

The cross-cohort transportability heatmap is shown in Figure 2.

Diagnostic model performance decayed below the prespecified threshold

Model performance showed the clearest evidence of limited diagnostic transferability. The primary gene-only model had AUC 0.875 in GSE63060 and 0.781 in GSE63061, but fell to 0.567 in GSE85426 and 0.554 in GSE140829. The primary integrated model had AUC 0.876 in GSE63060 and 0.784 in GSE63061, but fell to 0.568 in GSE85426 and 0.567 in GSE140829. A train-development refit model selected without final validation labels reached AUC 0.843 in combined GSE63060 plus GSE63061 training and AUC 0.587 in GSE85426 and 0.569 in GSE140829. No primary or pre-registered refit model achieved the prespecified independent cross-platform AUC threshold of 0.70.

The AUC drop from GSE63061 to cross-platform datasets exceeded 0.21 for both primary models. This supports the Route B interpretation that model performance is cohort-dependent and not consistently transferable across independent platforms. AUC decay and exploratory GSE140829 MCI score placement are shown in Figure 3. MCI score placement was intermediate between control and AD in 2 of 3 model summaries and was treated as exploratory support only.

Stable and unstable gene sets separated transportable from non-transportable signals

After the diagnostic model claim was downgraded, candidate genes were classified by transportability. The near-cohort directional anchor included 471 genes, and the near-cohort nominal anchor included 448 genes. The strict stable primary set contained 103 genes. A broader directional stable sensitivity set contained 327 genes. The unstable primary set

contained 235 genes, including 139 genes with cross-platform reversal and 190 genes with cross-platform signal loss. The strict stable and unstable primary sets had no gene overlap. These results show that a transportable component exists, but a large part of the discovery-derived signal is unstable across independent validation contexts.

Stable genes showed conservative pathway context, whereas unstable genes lacked coherent enrichment

The strict stable set showed significant pathway enrichment under the candidate-background analysis. It had 12 significant GO BP terms, 2 significant KEGG terms, and 52 significant Reactome terms at FDR < 0.05. Enriched modules were dominated by cytoplasmic translation, ribosomal biology, rRNA processing, and RNA-processing pathways. The directional stable sensitivity set retained Reactome support, with 45 significant Reactome terms and 2 significant KEGG terms, although GO BP support was weaker.

In contrast, the unstable primary set showed no FDR-significant GO BP, KEGG, or Reactome enrichment. These results suggest that the stable transportable subset has a coherent but conservative biological context, whereas non-transportable genes do not form a clear enriched pathway module. The pathway comparison is summarized in Figure 4.

Discussion

This study evaluated the cross-cohort transportability of AD peripheral blood transcriptomic signatures using a staged validation design. The central finding is that discovery-derived blood transcriptional signals can reproduce strongly in a related cohort but show limited diagnostic transferability across independent platforms and cohorts. This distinction is important because a related-cohort validation result alone could support an overly optimistic biomarker interpretation, whereas independent cross-platform validation reveals substantial degradation. The findings extend prior blood-transcriptomic AD studies by changing the evidentiary emphasis. Earlier work helped establish that blood expression profiles can contain AD- or ageing-related molecular signals and can yield apparently discriminative signatures in selected cohorts [5-8].

309 The present analysis does not dispute that premise. Instead, it shows that the same biomarker
310 domain can produce a strong related-cohort result and still fail a stricter cross-platform
311 diagnostic-transferability audit. This is a narrower but more defensible contribution for public-
312 data transcriptomics.

313 The GSE63061 results demonstrate that the discovery-derived signal is not simply random.
314 Mapping was complete, direction concordance was very high, hundreds of genes replicated at
315 $FDR < 0.05$, and model AUC exceeded 0.78. If the analysis had stopped at GSE63061, the
316 results could have been interpreted as evidence of an apparently promising blood transcriptomic
317 classifier. However, GSE63061 is a related validation cohort, and its strong performance must
318 be interpreted in the context of shared study structure and related platform characteristics.

319 The cross-platform datasets changed the interpretation. In GSE85426, FDR-level replication
320 disappeared and model AUC values remained below 0.59. In GSE140829, direction
321 concordance remained relatively high, but FDR-level replication and discrimination were weak,
322 and model AUC values remained below 0.57. The GSE140829 result is especially informative
323 because the dataset is large enough that failure cannot be dismissed only as a sample-size
324 problem. Instead, the result suggests that the measured blood transcriptional signal is sensitive
325 to platform, cohort composition, preprocessing, phenotype definition, or unmeasured biological
326 and technical factors.

327 These findings do not mean that peripheral blood transcriptomics is uninformative for AD.
328 Rather, they show that evidence standards matter. A signal can be biologically plausible and
329 reproducible in one external context while still lacking the stability required for diagnostic use.
330 For public-data biomarker studies, this is a critical distinction. External validation should include
331 cohorts that differ meaningfully from the discovery cohort; otherwise, model performance may
332 mostly reflect related-cohort reproducibility rather than clinical transferability.

333 The stable/unstable gene-set analysis adds biological context to the validation results. A strict
334 stable subset of 103 genes was identified, and this subset was enriched for translation,

335 ribosomal, and RNA-processing pathways. These pathways may represent a conservative
336 blood transcriptional background that is more transferable across cohorts than weaker or more
337 context-sensitive signals. However, this enrichment should not be overinterpreted as a disease-
338 specific AD mechanism. Many translation and RNA-processing terms are broad cellular
339 modules, and the current analysis cannot separate AD-related biology from blood-cell
340 composition, systemic physiology, RNA quality, or platform-related measurement stability. The
341 stable pathway signal is therefore best interpreted as conservative context for transportability,
342 not as evidence of brain AD pathogenesis.

343 The unstable gene set was also informative. It contained 235 genes that were nominally
344 reproducible in the related cohort but reversed or lost signal in cross-platform validation. This
345 set showed no FDR-significant GO BP, KEGG, or Reactome enrichment. The absence of
346 coherent enrichment supports the interpretation that unstable genes may reflect heterogeneous
347 cohort-sensitive signals rather than a single consistent biological program.

348 The diagnostic modeling results should be read as a cautionary finding. The primary models
349 performed well in training and related validation, but the same models failed to meet the AUC
350 0.70 threshold in GSE85426 and GSE140829. A refit strategy using GSE63060 plus GSE63061
351 also failed to rescue cross-platform performance. These results argue against presenting the
352 current signature as ready for clinical use. They instead support the need for strict validation,
353 transparent reporting of negative validation results, and claim audits before public transcriptomic
354 signatures are promoted as diagnostic biomarkers.

355 This study has limitations. First, it is based entirely on public transcriptomic datasets and lacks
356 wet-lab validation. Second, microarray platforms and annotation resources differ across cohorts,
357 which can affect mapping and expression comparability. Third, blood-cell composition could not
358 be fully adjusted across all datasets. Fourth, metadata harmonization was limited by the
359 information available in GEO records. Fifth, pathway enrichment provides biological context but

not causal mechanism. Sixth, MCI analyses in GSE140829 were exploratory and should not be treated as evidence of clinical staging performance.

Despite these limitations, the study provides a reproducible validation audit of AD blood transcriptomic signatures across multiple public cohorts. The results show that related-cohort reproducibility is achievable, but diagnostic transferability across independent platforms remains limited. Future work should combine standardized sample processing, cell-composition adjustment, independent prospective validation, and predefined reporting standards before peripheral blood transcriptomic signatures are advanced as diagnostic tools.

Conclusions

Peripheral blood transcriptional signatures of AD show strong related-cohort reproducibility but limited diagnostic transferability across independent platforms and cohorts. Stable transportable genes show conservative enrichment for translation and RNA-processing pathways, while unstable genes lack coherent pathway enrichment. These findings support cautious use of public blood transcriptomic signatures and argue for strict external validation before diagnostic biomarker claims.

List of abbreviations

AD: Alzheimer's disease; AUC: area under the receiver operating characteristic curve; FDR: false discovery rate; GEO: Gene Expression Omnibus; GO BP: Gene Ontology Biological Process; KEGG: Kyoto Encyclopedia of Genes and Genomes; MCI: mild cognitive impairment; ROC: receiver operating characteristic.

Declarations

Ethics approval and consent to participate

Not applicable. This study analyzed publicly available, de-identified datasets and did not involve new recruitment of human participants, new collection of human specimens, or access to directly identifiable private information.

385 Consent for publication

386 Not applicable.

387 Availability of data and materials

388 All datasets analyzed in this study are publicly available from GEO under accessions
389 GSE63060, GSE63061, GSE85426, and GSE140829. Analysis scripts are stored in the project
390 scripts/ directory. Main outputs are stored in results/, figures in figures/route_b/, and audit
391 reports in audit/. The supplementary reproducibility traceability table maps each main
392 manuscript claim to the corresponding script and result file. A public GitHub archive index and
393 manifest for the compact reproducibility package are available at
394 <https://github.com/1627626277-cyber/NO.2/tree/main/ad-blood-transcriptome-transportability>.
395 Raw GEO files and large intermediate expression matrices are not included in the compact
396 archive because they can be retrieved from GEO or regenerated from the scripts.

397 Competing interests

398 The author declares that there are no competing interests.

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401 Authors' contributions

402 Z.J. conceived the study, designed the analysis strategy, curated public datasets, performed the
403 computational analysis, interpreted the results, prepared the manuscript, and approved the
404 submitted version. This statement should be revised if additional authors are added before
405 submission.

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Figure Legends

Figure 1. Staged validation design for the Route B transportability analysis

The analysis used GSE63060 as the discovery cohort, GSE63061 as a related validation cohort, GSE85426 as a cross-platform stress-test cohort, and GSE140829 as a large independent cross-platform validation cohort. Candidate genes and model-selection rules were frozen before the cross-platform validation steps. The figure emphasizes the separation between discovery, related-cohort validation, and independent cross-platform validation.

Figure 2. Cross-cohort transportability audit of discovery-derived candidate genes

Heatmap showing mapping fraction, direction concordance, nominal replication fraction, FDR replication fraction, median gene-level oriented AUC, and best model AUC across validation cohorts. GSE63061 showed strong related-cohort replication, whereas GSE85426 and GSE140829 showed reduced FDR-level replication and model AUC values below 0.70.

Figure 3. Diagnostic AUC decay and exploratory MCI score placement

The upper panel summarizes AUC values across training, related validation, and cross-platform validation contexts for the primary gene-only model, primary integrated model, and train-development refit model. The dashed line indicates the prespecified AUC threshold of 0.70. The lower panel shows GSE140829 score distributions for Control, MCI, and AD samples using the frozen primary gene-only signature. MCI placement is exploratory and was not used for model optimization.

Figure 4. Pathway context of stable and unstable transportability classes

The figure compares FDR-significant GO BP, KEGG, and Reactome enrichment yields for the strict stable and unstable gene sets. Stable transportable genes showed significant enrichment dominated by ribosomal translation and RNA-processing pathways, whereas unstable genes did not show FDR-significant enrichment in GO BP, KEGG, or Reactome under the candidate-background analysis.

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Figures

Staged Route B validation design



Figure 1. Staged validation design for the Route B transportability analysis.



Figure 2. Cross-cohort transportability audit of discovery-derived candidate genes.

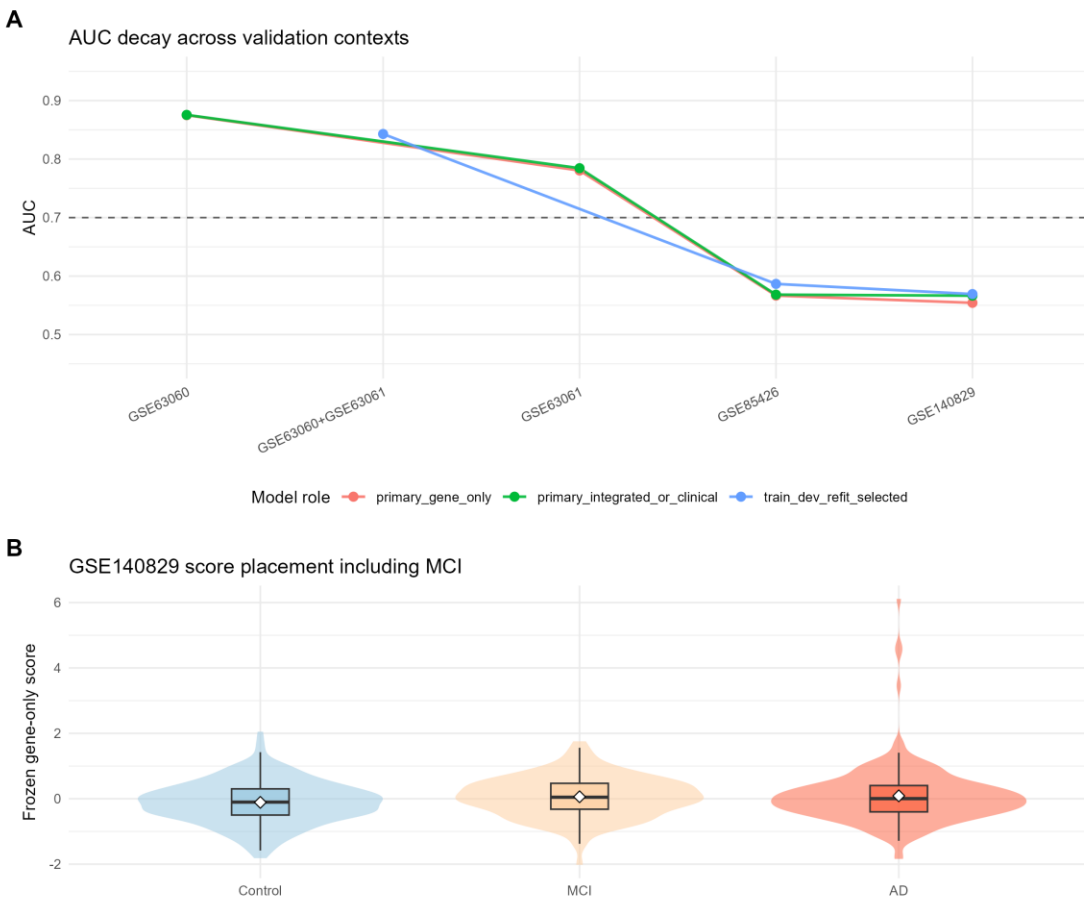


Figure 3. Diagnostic AUC decay and exploratory MCI score placement.

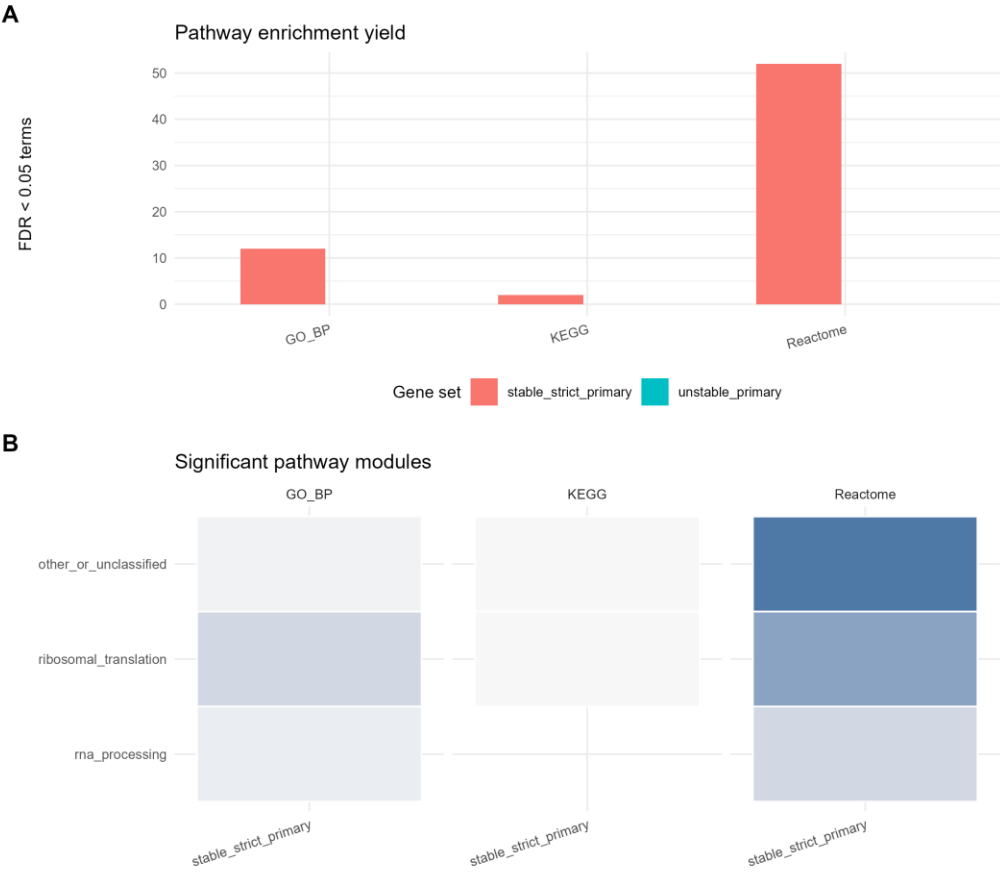


Figure 4. Pathway context of stable and unstable transportability classes.